

Incident handler's journal

Date: May 5, 2025	Entry: #1
Description	The cause of the security incident was a phishing email that contained a malicious attachment. Once it was downloaded, ransomware was deployed, encrypting the organization's computer files. An organized group of unethical hackers left a ransom note stating that the company's files were encrypted and demanding money in exchange for the decryption key Documenting a cybersecurity incident This incident occurred in the two phases: Detection and Analysis: The scenario outlines how the organization first detected the ransomware incident. For the analysis step, the organization contacted several organizations for technical assistance. Containment, Eradication, and Recovery: The scenario details some steps that the organization took to contain the incident. For example, the company shut down their computer systems. However, since they could not work to eradicate and recover from the incident alone, they contacted several other organizations for assistance.
Tool(s) used	None
The 5 W's	• Who: An organized group of unethical hackers • What: A ransomware security incident • When: This incident occurred on Tuesday at 9:00 a.m. • Where: At a health care company • Why: This happened because a group of unethical hackers were able to gain access to the company's systems through a phishing email that

	contained a malicious attachment was sent to the employees of the clinic. After gaining access, ransomware was deployed on the company's, encrypting the organization's critical files. The attackers' motivation appears to be financial because the ransom note they left demanded a large sum of money in exchange for the decryption key.
Additional notes	The health care company can provide best security practice seminars to inform employees of the proper way to avoid and prevent social engineering attacks. The company should contact a third party security company to come restore their systems and eradicate the malware.

Date: May 5, 2025	Entry: #2
Description	Investigate a suspicious file hash
Tool(s) used	VirusTotal, which is an investigative tool that analyzes files and URLs for malicious content such as viruses, worms, trojans, and more. SHA256 file hash
The 5 W's	• Who: An unknown malicious actor • What: An email sent to an employee contained a malicious file attachment with the SHA-256 file hash of 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b • When: Occurred from 1:11 pm to 1:20 pm • Where: An employee's computer at a financial services company

	Why: A threat actors phished an employee at the company, and successful got them to download and execute a file containing malware
Additional notes	The file hash from the file that was downloaded by the employee has been reported as malicious by over 50 vendors. This file hash is known as the malware Flagpro, which has been commonly used by the advanced threat actor BlackTech. This incident can be prevent in the future by ensuring employees have gone through security awareness training so that employees are careful with what they click on.

Date: May 5, 2025	Entry: #3
Description	Used a playbook to respond to a phishing incident
Tool(s) used	None
The 5 W's	- Who: Threat actor - What: A user opened a malicious email and opened attachments or clicked a link. - When: Occurred at 2:00pm - Where: On a company laptop, within the company office. - Why: This happened because an employee fell victim to a phishing attack, conducted by a threat actor.

Date: May 05, 2025	Entry: #4
Description	Reviewing a final incident report
Tool(s) used	None.
The 5 W's	• Who: A malicious actor • What: An individual was able to gain unauthorized access to customer personal identifiable information (PII) and financial information • When: December 28, 2022, at 7:20 p.m. • Where: On the e-commerce web app • Why: The root cause of the incident was identified as a vulnerability in the e-commerce web application
Additional notes	The final report does well in covering all aspects of the incident. And clearly depicts timeline, investigation, response, remediation, and future recommendations.

Date: May 14, 2025	Entry: #5
Description	Analyzing a packet capture file
Tool(s) used	For this activity, I used Wireshark to analyze a packet capture file. Wireshark is a network protocol analyzer that uses a graphical user interface. The value of Wireshark in cybersecurity is that it allows security analysts to capture and analyze network traffic. This can help in detecting and investigating malicious activity.
The 5 W's	• Who: N/A • What: N/A • Where: N/A

	• When: N/A • Why: N/A
Additional notes	At first glance, the interface was very overwhelming as there is lots of information being displayed. I can see why it's such a powerful tool for understanding network traffic.

Date: May 14, 2025	Entry: #6
Description	Capturing packets with tcpdump.
Tool(s) used	For this activity, I used tcpdump to capture and analyze network traffic. Tcpdump is a network protocol analyzer that's accessed using the command-line interface. Similar to Wireshark, the value of tcpdump in cybersecurity is that it allows security analysts to capture, filter, and analyze network traffic.
The 5 W's	• Who: N/A • What: N/A • Where: N/A • When: N/A • Why: N/A
Additional notes	Since I have practice and experience with the linux command line I found using Tcpdump natural. And by carefully following the instructions, I was able to get through this activity and capture network traffic.

Date: May 14, 2025	Entry: #7
Description	This activity involved exploring Suricata , an open-source IDS/IPS tool, to understand how it monitors network traffic. The focus was on the general process of rule creation, triggering alerts based on specific network

	packets, and analyzing the resulting log outputs to interpret security events.
Tool(s) used	Suricata: Used for intrusion detection, intrusion prevention, and network analysis.
The 5 W's	• Who: N/A • What: N/A • Where: N/A • When: N/A • Why: N/A
Additional notes	Suricata relies on custom-written rules to define which traffic to process. Log Files: Two primary log files were emphasized: • fast.log: Likely used for quick identification of triggered alerts. • eve.json: A more detailed, structured log format used for deeper analysis and integration with other tools. Configuration: Proper source and destination network specifications in the Suricata configuration are essential for accurate monitorin

Reflections/Notes:

After taking this course, my understanding of incident detection and response has definitely evolved. At the beginning of the course, I had some basic understanding of what detection and response entailed, but I didn't fully understand the complexity involved. As I progressed through the course, I learned about the lifecycle of an incident; the importance of plans, processes, and people; and tools used. Overall, I feel that my understanding has changed, and I am equipped with more knowledge and understanding about incident detection and response.